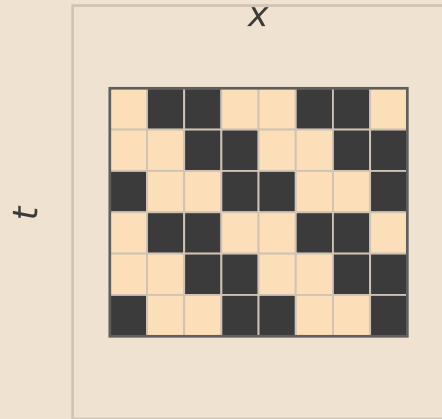


Relative-periodic domain selection by Bernoulli NML

1 Observed spacetime

Choose candidate (p, s) .
Example: $p = 2, s = 1$.



Raw CA spacetime to explain.

2 Orbit classes

Apply $(t, x) \mapsto (t + p, x + s)$.
Same color = same orbit.

1	2	3	4	1	2
4	1	2	3	4	1
3	4	1	2	3	4
2	3	4	1	2	3
1	2	3	4	1	2

Each orbit contributes one background bit.

3 Majority-vote fit

Count ones and zeros in each orbit.
Choose the majority bit.

orbit	ones / zeros
1	5 / 1
2	1 / 5
3	4 / 2

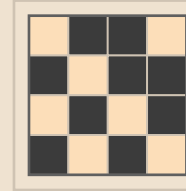
Output
one bit per orbit

Residual mask: $M = U \oplus B^*$.

4 Score and select

NML = fit + penalty.
Keep the best period.

$B^*(p, s)$



Fit $\sum_j n_j H_b(\hat{\theta}_j)$

Penalty $\frac{1}{2} \sum_j \log_2 n_j$

Select
Lowest NML wins.

Output: periodic background + residual.